FORENSIC eDNA CHAIN OF CUSTODY & LAB TRANSMITTAL FORM

PROJECT REVVEL / GROUNDTRUTH AR — FORENSIC GENETICS DIVISION

Submitting Agency: GroundTruth AR Cold Case Unit	**Transmittal Date (UTC):** 2026-10-10 16:30	**Target Protocol:** MPS / eDNA GRCh38 Alignment
Lead Investigator: Audrey Evans (Data Architect)	**Destination Lab:** Accredited Forensic Genetics Lab	**Transport Temp:** Cold Storage (-20°C / Dry Ice)

1. SAMPLE INVENTORY & PCR INHIBITION NEUTRALIZATION DOSAGE

Sample ID	Case ID & Target	GPS Coordinates	Soil Matrix	Depth	BSA / Milk Dosage	Kit & Incubation Protocol
SOIL-LAB-001	SHAFFER-2006-OH Columbus Pearl Alley	39.9985° N -83.0064° W	Urban Silt-Soot Vault Bed	0.15 m (15 cm)	BSA: 12.0 mg/mL Milk: 0.15 g/g	Silica-Column PowerSoil + Hydrocarbon Wash Proteinase K Digestion @ 56°C (30 min)
SOIL-LAB-002	SHTUKA-2018-BC Fairways Creek Log Jam	50.8825° N -119.8851° W	Alpine Till & Volcanic Ash	0.10 m (10 cm)	BSA: 12.0 mg/mL Milk: 0.20 g/g	Qiagen PowerSoil Pro + BSA Spike Thermal Incubation @ 65°C (10 min)
SOIL-LAB-003	LINDEMAN-2026-MO Dogwood Ravine Outfall	38.4512° N -91.8234° W	Alluvial Lowland Silt-Clay	0.08 m (8 cm)	BSA: 10.0 mg/mL Milk: 0.10 g/g	PowerSoil Pro + Double Wash Incubate w/ C1 Buffer @ 60°C
SOIL-LAB-004	FARBER-2015-PA Sharp Mountain Coal Bench	40.6781° N -76.1923° W	Coal Spoil Bench	0.20 m (20 cm)	BSA: 12.0 mg/mL Milk: 0.15 g/g	PowerSoil Pro + Iron-Chelating Wash Thermal Incubation w/ BSA/Milk @ 65°C

2. CHAIN OF CUSTODY TRANSFER REGISTER

Item # / Sample ID	Relinquished By (Print & Sign)	Received By (Print & Sign)	Date / Time (UTC)	Reason for Transfer / Transport Method	Tamper Seal Intact?
SOIL-LAB-001 SOIL-LAB-002	A. Evans, Lead Investigator *A. Evans*	Secured Courier Specialist *J. Doe*	2026-10-10 16:30	Transport to Forensic Genetics Lab via Cold Storage Unit	[X] YES [ ] NO
SOIL-LAB-003 SOIL-LAB-004	A. Evans, Lead Investigator *A. Evans*	Secured Courier Specialist *J. Doe*	2026-10-10 16:30	Transport to Forensic Genetics Lab via Cold Storage Unit	[X] YES [ ] NO
SOIL-LAB-001 - 004	Secured Courier Specialist _____	Lab Intake Officer _____ _____	____-____-____ ____:____	Laboratory Intake & Cold Storage Ingestion (-20°C)	[ ] YES [ ] NO
SOIL-LAB-001 - 004	Lab Intake Officer _____ _____	DNA Analyst / Technologist _____	____-____-____ ____:____	Extraction & Competitive Inhibitor Neutralization Run	[ ] YES [ ] NO

3. LABORATORY TECHNICAL INSTRUCTIONS & EVIDENCE WITNESS HASH

LABORATORY EXTRACTION PROTOCOL MANDATE:

1. **Competitive Inhibitor Blocking Buffer:** Prior to cell lysis, add 10.0–12.0 mg/mL Bovine Serum Albumin (BSA Fraction V) and 0.05–0.20 g/g non-fat skim milk powder to bind humic/fulvic acids and chelate pyritic iron oxides/petroleum hydrocarbons.
2. **Incubation & Washing:** Execute sample-specific incubation (e.g., 56°C Proteinase K or 65°C thermal spike) as detailed in Section 1.
3. **Sequencing Alignment:** Perform Massively Parallel Sequencing (MPS) and align extracted reads against human reference genome **GRCh38**.

4. **Verification Threshold:** Cross-reference composite optical/LIF fluorescence scores. A match score exceeding 98.0% indicates deterministic soil-DNA alignment.

SHA-256 Cryptographic Evidence Witness Hash:
c1c433ff4d1efc47d60bfa26f38a5970d5c686a1ef5674319b823d1f6dd63474